ModelIt!

Teacher's Guide

Why Do You Weigh Less on the Moon?

Gravity, Mass, and the Force That Keeps You Grounded

Quick Reference

Grade Level	Grade 07L2
NGSS Standard	MS-PS2-4, MS-ESS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: MS-PS2-4, MS-ESS1-2

Construct and present arguments using evidence to support the claim that gravitational interactions are attractive and depend on the masses of interacting objects. Develop and use a model to describe the role of gravity in the motions within galaxies and the solar system.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model to describe how gravitational pull depends on planet mass and distance, and how this affects object weight and motion.
Disciplinary Core Idea	**PS2.B Types of Interactions / ESS1.B Earth and the Solar System** — Gravitational forces are always attractive and depend on the masses of interacting objects. Gravity governs the motions of objects in the solar system.
Crosscutting Concept	**Systems and System Models** — Students model the gravitational system with inputs (planet mass, distance) and outputs (gravitational pull, weight, jump height) to understand how the system behaves under different conditions.

CAST Testing Alignment

MULTIPLE CHOICE:

An astronaut weighs 180 pounds on Earth. On the Moon, where gravity is about 1/6 of Earth's, what would the astronaut weigh, and would their mass change?

CONSTRUCTED RESPONSE:

Using your gravity model, explain why Jupiter's moon Europa (which has less mass than Earth's Moon) would give you a different weight than standing on Jupiter itself. Describe how both planet mass and distance from center affect the gravitational pull you experience.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Gravity is the universal force of attraction between all objects with mass. It is what gives you weight, keeps the Moon orbiting Earth, and holds the solar system together. Understanding gravity means understanding the difference between mass and weight, and how the gravitational pull of any body depends on its mass and your distance from its center.

Mass vs. Weight

Mass is the amount of matter in an object and is measured in kilograms. It never changes regardless of location. Weight is the force of gravity acting on that mass and is measured in newtons (or pounds). Your mass is the same on Earth, the Moon, and Jupiter, but your weight is different on each because the gravitational pull is different.

What Determines Gravitational Pull

Gravitational pull depends on two factors: the masses of the objects involved and the distance between their centers. More massive planets pull harder. Greater distance means weaker pull. This follows the inverse square law: if you double the distance, the gravitational force drops to one-quarter.

Gravity in the Solar System

The Sun's massive gravity (333,000 times Earth's mass) holds all planets in orbit. Closer planets orbit faster because they experience stronger gravitational pull. Moons orbit planets for the same reason. Even you and Earth are gravitationally attracted to each other, but Earth is so much more massive that you move toward it, not the other way around.

Gravity and Space Exploration

Understanding gravity is essential for space travel. Rocket launches must overcome Earth's gravitational pull. Spacecraft use gravitational slingshots around planets to gain speed. Astronauts on the ISS experience microgravity (not zero gravity) because they are in free fall around Earth, moving fast enough sideways that they keep missing it.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify planet mass, distance from center, gravitational pull, object weight, and jump height as components of the gravity system.

E **Establish Relationships**
Students determine that planet mass and distance are external inputs that drive gravitational pull, which in turn determines weight and jump height.

V **Visualize & Model**
Students build a model showing how changing planet mass and distance affects gravitational pull and its consequences for objects on the surface.

E **Evaluate Outcomes**
Students run Moon, Jupiter, and orbital distance scenarios to observe how gravitational pull, weight, and jump height respond to different conditions.

R **Revise & Extend**
Students add atmospheric density or orbital velocity to explore how gravity interacts with other factors in planetary environments.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"If I put you on a scale right now, you'd get one number. But what if I put you on a scale on the Moon? Would the number change? Why?"*

Watch For

Have students predict their Moon weight. Use a quick calculator: Earth weight / 6 = Moon weight.

Slide 2: Learning Objectives

3 min

Say: *"Today we're going to figure out why gravity changes from place to place in the solar system."*

Watch For

Have students read objectives. Pre-teach mass vs. weight distinction.

Slide 3: Big Question

3 min

Say: *"Your body doesn't change when you go to the Moon. Same bones, same muscles, same atoms. So why does the scale say something different?"*

Watch For

Show astronaut footage from Apollo missions with bouncing lunar walks. Discuss.

Slide 4: LEVER Framework

2 min

Say: *"We'll build a model of how gravity works across the solar system."*

Watch For

Preview LEVER steps. Show comparative gravity table for planets.

Slide 5: Activity 1: Components

8 min

Say: *"What factors affect gravity? Which ones can we change in our model, and which ones respond?"*

Watch For

Guide sorting. Discuss why planet mass and distance are external driving inputs.

Slide 6: Activity 2: Connections

8 min

Say: *"When planet mass goes up, what happens to gravitational pull? When distance increases, what happens?"*

Watch For

Help students trace positive and negative relationships. Introduce inverse square concept.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's visit the Moon, then Jupiter, then float out into deep space. Watch what happens to your weight."*

Watch For

Students predict first, then run all three scenarios. Calculate actual weights on each body.

Slide 8: Discoveries

5 min

Say: *"So what's the real answer? Is it YOU that changes, or is it the PLACE that changes? What did our model prove?"*

Watch For

Lead discussion reinforcing mass vs. weight. Connect to orbital mechanics.

Slide 9: STEM Challenge

12 min

Say: *"NASA needs your help. How do you keep astronauts healthy when gravity is working against you?"*

Watch For

Groups design fitness programs using gravity data from their models.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Planet Mass, Distance from Center
- **INTERNAL:** Gravitational Pull, Object Weight, Jump Height

Relationships

- **Planet Mass to Gravitational Pull:** POSITIVE (+) — Greater planet mass means stronger gravitational pull. A more massive planet attracts objects with more force.
- **Distance from Center to Gravitational Pull:** NEGATIVE (-) — Greater distance from the planet's center means weaker gravitational pull. Gravity decreases with the square of the distance.
- **Gravitational Pull to Object Weight:** POSITIVE (+) — Stronger gravitational pull means greater object weight. Weight is the force of gravity acting on mass.
- **Gravitational Pull to Jump Height:** NEGATIVE (-) — Stronger gravitational pull means lower jump height. More gravity holds you down, making it harder to leave the surface.
- **Object Weight to Jump Height:** NEGATIVE (-) — Greater object weight (due to stronger gravity) means lower jump height, because more force is needed to overcome the downward pull.

Key Discoveries

- Weight changes depending on where you are, but mass always stays the same because mass is the amount of matter in your body
- Gravitational pull depends on BOTH the planet's mass AND your distance from its center, not just one factor
- The Moon has about 1/6 of Earth's surface gravity, so you would weigh 1/6 as much and could jump 6 times higher
- Gravity is what keeps planets in orbit around the Sun and moons in orbit around planets, because gravitational interactions are always attractive

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"There's no gravity in space"	There IS gravity everywhere in space. The ISS orbits at an altitude where gravity is about 90% as strong as on Earth's surface. Astronauts float because they're in free fall, not because gravity is absent. Gravity extends infinitely; it just gets weaker with distance.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Heavier objects fall faster"	In a vacuum (no air resistance), ALL objects fall at the same rate regardless of mass. A feather and a bowling ball dropped on the Moon (no air) hit the ground at the same time. This was actually demonstrated by Apollo 15 astronaut David Scott.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Weight and mass are the same thing"	Mass is how much matter you contain (measured in kg) and never changes. Weight is the gravitational force on that mass (measured in N or lbs) and changes depending on where you are. A 70 kg person has 70 kg of mass on Earth, the Moon, Jupiter, or floating in deep space, but their weight is different in each location.